

MAX ZHU

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S. in Computer Engineering and Mathematics, Minor in Robotics

Expected Graduation: May 2027

- **Concentrations:** Systems & Architecture, Computing Hardware & Emerging Architecture **GPA: 3.91/4.00**
- **Relevant Coursework:** Embedded Systems Design, HW/SW Programming, Operating Systems, Circuit Analysis, Digital Signal processing, Data Structures and Algorithms, Object Oriented Programming, VLSI and Advanced Digital Design

SKILLS

Software: C++, C, Python, Java, RISC-V, FreeRTOS, Perl, Linux, MATLAB, Django, SQL, HTML, Git

Hardware: FPGA, ESP32, SystemVerilog, Oscilloscopes, UART, I2C, SPI, UVM, GTKWave, CAD

EXPERIENCE

FUSMobile

Apr 2025 – Aug 2025

Embedded and QA Engineer

Alpharetta, GA

- Engineered a bidirectional USB serial protocol between an Arduino Due and an imaging workstation to enable closed-loop automatic self-alignment, parsing live X-ray inputs from the Neurolyzer to drive real-time spatial orientation
- Built an automated medical report generation system in C++, using MFC for metadata extraction, OpenCV for image filtering/processing, and libHaru for annotation, reducing manual reporting time by 90%
- Re-engineered the Neurolyzer, increasing portability by 20% and interactivity, for hands-on conference demonstrations

MARGA Lab

Aug 2024 – May 2025

Wireless Digital Twin Researcher

Atlanta, GA

- Created Python scripts with Sionna to optimize transmitter-receiver placement by over 60% in over 50 unique environments
- Analyzed and annotated 30+ research papers on 5G, Wi-Fi 6, ray tracing, and UAVs to establish baseline models
- Published and presented a paper on varying geometric simplifications applied to Blender environments for WDTs, reducing simulation rendering time by 50% while preserving over 95% signal fidelity compared to full-scale models

Silicon Jackets

Aug 2024 – Present

Digital Design and Verification Engineer

Atlanta, GA

- Driving RTL development and verification across two tapeouts for a 100MHz RV32I ALU, shifter, and L1 data-cache pipeline
- Authoring micro-architectural specifications and block diagrams, translating system requirements into synthesizable SystemVerilog
- Established a robust verification infrastructure using a UVM env (agents/scoreboard) and an ISS reference to achieve 100% functional and $\geq 95\%$ code coverage, revealing 10 edge cases through automated Python/CI regression pipelines

PROJECTS

ESP32 Embedded Systems Projects | ESP32-C6, C++

- Developed multiple embedded applications on the ESP32 using memory-mapped I/O to interface with hardware peripherals including digital inputs/outputs, analog signals, actuators, UART, SPI, and I2C devices
- Built a lock screen integrating a keypad (MPR121 over I2C) and IMU sensor (ICM-20948 over I2C) with input parsing for passcode entry, implementing a state-machine driven authentication system with real-time feedback on a uLCD display (over UART)
- Created a live Spotify hardware controller using the Spotify Web API, enabling physical playback control (play, pause, volume) and live data retrieval (album cover and playback bar), the project gained directly led to being selected as a teaching assistant

XV6 Kernel | C, C++, RISC-V, Perl

- Implemented kernel level threads by adding the clone system call and user space mutexes, manually manipulating RISC-V page tables to support up to 64 concurrent threads per process within a shared virtual memory space
- Enforced user isolation by extending the file system with 9-bit POSIX read/write/execute permissions, efficiently modifying the 64-byte inode structure to validate user credentials without increasing disk sector overhead
- Changed the scheduler from Round-Robin to a 3-level Priority-Based queue, reducing wait times for high-priority tasks by 15%, and rewrote the bootblock assembly to support kernel images larger than the default 512-byte limit

Custom Macro Mechanical Keypad | C, USB Protocol (TinyUSB), KiCAD, QMK

- Designed a USB Human Interface Device capable of emulating keyboard and mouse actions using endpoint descriptors
- Wrote a matrix scanning algorithm with software debouncing (5ms integration window) to handle a 4x4 switch matrix
- Designed and routed a custom 2-layer PCB in KiCad, integrating USB-C ESD protection and differential pair routing

EXTRACURRICULARS

Leadership: GT IEEE (Director of Corporate Affairs), Hive (Master Peer Instructor), ECE Student Ambassadors

Clubs: Silicon Jackets (Digital Designer), Chinese Student Association (Dynasty Head), Trading@GT (Prediction Markets), Phi Delta Theta

Interests: Motorcycles, Weightlifting, Algorithmic Trading, Cars, Snowboarding, Keyboards, Photography